EXERCISE 1
SQL FUNDAMENTALS

1. Write a SQL query to retrieve all columns from employees table

Syntex
```
SELECT *
FROM `workspace`.`default`.`employees`;
```

RESULTS

ID	First_Name	Last_Name	Department	Salary	Hire_Date	City
1	~~Sarah~~ John	Doe	IT	55000	2018/06/15	New York
2	Jane	Smith	HR	48000	2019/07/20	Chicago
3	Mike	Johnson	Finance	60000	2017/09/30	Los Angeles
4	Sarah	Brown	IT	53000	2021/03/25	New York
5	David	White	Marketing	52000	2016/04/10	San Francisco
6	Emily	Davis	IT	62000	2015/02/14	Chicago
7	Robert	Wilson	Finance	59000	2019/10/01	Houston
8	Jessica	Moore	HR	51000	2018/05/22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago
10	Laura	Hall	IT	50000	2020/08/10	San Francisco

2. Write a SQL query to find all the unique departments in the employees table

Syntex
```
SELECT DISTINCT department
FROM `workspace`.`default`.`employees`;
```

RESULTS

Department
IT
HR
Finance
Marketing

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3. Write a SQL query to retrieve all employees first and last names ordered by salary in descending order

Syntex
```
SELECT First_name, last_name, salary,
FROM 'workspace'.'default'.'employees'
ORDER BY salary DESC;
```

Results

first_Name	last_Name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

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4. Write a SQL query to retrieve the top 5 highest-paid employees

Syntax

```
SELECT first_name, lastname, salary
FROM 'workspace'.'default'.'employees'
ORDER BY Salary DESC
limit 5;
```

Results

first_name	last_name	Salary
Emily	Davis	62000
Mike	Johnson	60 000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000

5. Write a SQL query to find employees who work in IT department

Syntax

```
SELECT *
FROM 'workspace'.'default'.'employees'
WHERE department = 'IT';
```

Results

ID	first_name	lastname	department	Salary	Hire date	City
1	John	Doe	IT	55000	2018/06/15	New York
4	Sarah	Brown	IT	53 000	2021/03/25	New York
6	Emily	Davis	IT	62000	2015/02/14	Chicago
10	Laura	Hall	IT	50 000	2020/08/10	San Fransico

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6. Write query to find employees who work in the finace dep
AND have a salary greater than 58000

Syntex
```
SELECT *
FROM 'workspace'.'default'.'employees'
WHERE department = 'Finance' And Salary > 58000;
```

Results

ID	first_name	last_name	department	Salary	Hire_date	City
3	Mike	Johnson	Finance	60000	2017/09/30	Los Ang
7	Robert	Wilson	Finance	59000	2018/10/01	Houst

7. Write a query to find employees who work in HR department
OR the marketing department

Syntex
```
SELECT first_name, last_name, department
FROM 'workspace'.'default'.'employees'
WHERE department = 'HR' OR department = 'marketing';
```

Results

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

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8. Write a SQL query to find employees who do not work in the IT department

Syntex
```
SELECT first_name, last_name, department
FROM `workspace`.`default`.`employees`
WHERE department != 'IT';
```

Results

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. Write a SQL query to find employees who are in HR, IT, or Finance department

Syntex
```
SELECT first_name, last_name, department
FROM `workspace`.`default`.`employees`
WHERE department IN ('IT', 'HR', 'Finance');
```

Results

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

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10. Write a query to find employees who are in the IT department, have a salary greater than 50000, and are located in New York

Syntex
```
SELECT firstname, lastname, department, Salary, City
FROM 'workspace'.'default'.'employees'
WHERE department = 'IT' AND Salary > 50000
AND city = 'New York';
```

Results

first_name	last_name	department	Salary	City
John	Doe	IT	55000	New York
Sarah	Brown	IT	53000	New York

11. Write a SQL query to retrieve the first and last names of employees who work in the Finance or Marketing department, earn more than 52000 and order the results by salary in descending order

Syntex
```
SELECT first_name, lastname, department, salary
FROM 'workspace'.'default'.'employees'
WHERE department = 'Finance' OR department = 'Marketing'
AND salary > 52000
ORDER BY salary DESC;
```

first_name	last_name	department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

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12. Write a SQL query to find all unique ~~employees~~ cities where employees work, excluding those in the IT and HR department

```
SELECT DISTINCT city, department
FROM `workspace`.`default`.`employees`
WHERE department NOT IN ('IT','HR');
```

Results

City	department
Los Angeles	Finance
San Franscisco	Marketing
Houston	Finance
Chicago	Marketing

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13. Write a SQL query to retrieve employees who are not in the finance department, have a salary greater than 50000, and order the results by hire date in ascending order

Syntax
```
SELECT first_name, last_name, department, salary, hire_date
FROM `workspace`.`default`.`employees`
WHERE department != 'Finance'
AND Salary > 50000
ORDER BY hire_date ASC;
```

Results

first_name	last_name	department	Salary	hire_date
Emily	Davis	IT	62000	2015/02/14
David	White	Marketing	52000	2016/04/10
Jessica	Moore	HR	51000	2018/05/22
John	Doe	IT	55000	2018/06/15
Sarah	Brown	IT	53000	2021/03/25
Daniel	Clark	Marketing	53000	2022/06/01

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14. Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department.

Syntex

```
SELECT *
FROM `workspace`.`default`.`employees`
WHERE City IN ('Chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
Limit 3;
```

Results

ID	first_name	last_name	department	Salary	Hiredate	City
6	Emily	Davis	IT	62000	2015/02/14	Chicago
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago

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